

Payment & Financial Services - Document 15 - Page 1/6

Document ID: PAY-015 | Classification: Internal Use

Payment fraud detection: velocity checks, device fingerprinting, and ML-based risk scoring.

Refund must be $\leq$ original amount. Partial refunds supported. Full refund voids original transaction.

Payment webhooks: event types, signature verification, retry policy, and idempotency handling.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 15 - Page 2/6

Document ID: PAY-015 | Classification: Internal Use

Merchant account management: onboarding, KYC verification, and settlement schedule configuration.

3DS2 challenge: customer may receive OTP from bank. Timeout 5 minutes. Handle challenge completion.

Financial reporting: transaction reports, revenue analytics, and regulatory reporting formats.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 15 - Page 3/6

Document ID: PAY-015 | Classification: Internal Use

Payment webhooks: event types, signature verification, retry policy, and idempotency handling.

Settlement cycle: T+2 for cards, T+1 for bank transfer. Check processor-specific timelines.

Payment SLA and uptime: 99.9% target, incident response times, and status page communication.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 15 - Page 4/6

Document ID: PAY-015 | Classification: Internal Use

Financial reporting: transaction reports, revenue analytics, and regulatory reporting formats.

Payment reconciliation: match transaction ID, amount, currency. Flag mismatches for manual review.

Error code 5003: Payment gateway connection timeout. Check network connectivity and gateway health.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 15 - Page 5/6

Document ID: PAY-015 | Classification: Internal Use

Payment SLA and uptime: 99.9% target, incident response times, and status page communication.

Merchant KYC: business registration, bank account verification, beneficial owner disclosure.

Error code 4001: Invalid payment method. Verify card details and expiry. Support 3DS if required.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

Payment & Financial Services - Document 15 - Page 6/6

Document ID: PAY-015 | Classification: Internal Use

Error code 5003: Payment gateway connection timeout. Check network connectivity and gateway health.

Subscription billing: prorate on upgrade/downgrade. Credit unused portion. Send invoice 7 days before charge.

Error code 4002: Insufficient funds. Customer must use alternative payment method or add funds.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.